

A Post-Quantum Privacy-Enhanced Federated Learning Model for Driver Behavior Profiling

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Abstract—As vehicle systems become increasingly connected and intelligent, insurance providers are turning to machine learning techniques to personalize billing based on individual driving behavior. This shift raises important questions about how to balance predictive performance with user privacy. In this paper, we present PrivFedProfiling, a decentralized privacy-preserving learning framework designed for use-based insurance (UBI) systems. Our method leverages Federated Learning (FL) to collaboratively train behavior models across distributed driver devices without transferring raw data. To further strengthen privacy, we integrate Differential Privacy (DP) and Homomorphic Encryption (HE) within the training process, protecting sensitive patterns in shared model updates. The proposed approach uses a Multilayer Perceptron (MLP) architecture and is validated using synthetic driving behavior data generated from the SUMO simulator. It offers a realistic yet controllable environment for testing. Results indicate that our method maintains high model accuracy while ensuring strong privacy guarantees, making it suitable for real-world deployment.

Index Terms—Car insurance, Private machine learning construction, privacy-preserving, SUMO simulations.

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I. INTRODUCTION

The rise of intelligent transportation systems is reshaping driver-vehicle interaction and behavior monitoring. Among these innovations, Usage-Based Insurance (UBI), especially Pay-As-You-Drive (PAYD), offers a fairer, behavior-linked pricing model by using real-time vehicle data to promote safer, eco-friendly driving. There is increasing demand for usage-based insurance models that reward safe driving, but they rely on collecting sensitive real-time data, often without sufficient privacy protections. However, this type of data is inherently sensitive and is often processed by insurance companies without adequate access controls or privacy safeguards. Several of the associated privacy risks in connected vehicles have been identified and discussed in our previous work [1], [2]. To avoid these privacy threats, the data has to comply with privacy regulations such as the General Data Protection Regulation (GDPR) and the California Consumer Privacy Act (CCPA) [3]. Insurance systems must be designed with privacy-by-design principles. This includes ensuring unlinkability, anonymity, plausible deniability, undetectability, and confidentiality, as detailed in prior work [1].

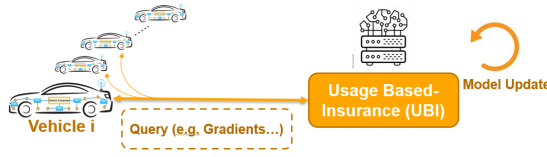


Fig. 1. ML Construction for UBI (Training part)

Artificial Intelligence (AI) offers significant benefits for *Usage-Based Insurance* (UBI), particularly in enhancing *driver profiling*. By processing large volumes of driving data, AI systems—especially those based on *Machine Learning* (ML)—can classify drivers into behavioral categories (e.g., Slow, Normal, Dangerous) and continuously improve through adaptive learning³. Besides the well-known benefits of ML decision-making, automation and efficiency, adaptability, and advanced data analysis—privacy-preserving ML training still faces critical challenges. These include: **Privacy Preservation, Scalability, Performance, Communication Overhead, and Practical Feasibility** (see more in Section II). In addition, effective driver profiling requires initial training on *sensitive personal data*, including location, speed, acceleration, and direction. To address these issues, we propose *Privacy-Preserving Federated Profiling Model (PrivFedProfiling)*, a decentralized approach for UBI driven by behavioral classification. The main contributions of this work are summarized as follows:

- **Proposal of PrivFedProfiling:** We propose a decentralized ML algorithm for UBI services that classifies driving behavior (slow, normal, dangerous), validated using SUMO-simulated data.
- **Privacy Protection:** A protocol combining Differential Privacy (DP) and Homomorphic Encryption (HE) is designed to ensure data confidentiality and GDPR compliance.
- **Implementation and Evaluation:** We implement the full pipeline and conduct a performance evaluation, showing model stability with and without the cryptographic layer. Code and datasets are available on GitHub⁴.
- **New SUMO-Based Dataset:** We generate a novel dataset with SUMO, comprising 88 vehicle profiles (29 slow, 29 normal, 30 dangerous),

³Source: <https://medium.com/@careervira.community/most-important-advantages-and-disadvantages-of-machine-learning-in-2024-9d56ee451622>

⁴Our Github: https://github.com/badr007-01/Usage_Based_insurance_DP_HE_from_SUMODataset.git

each with 90–120 trips.

To the best of our knowledge, this is the first work to propose a federated learning framework for UBI integrating HE and DP for driver privacy protection.

II. RELATED WORK

To comply with GDPR principles in driver profiling, various cryptographic systems aim to protect vehicle-sensitive data and ensure the private retrieval of information. Some works propose local computation solutions, such as PriPAYD [4], a pay-as-you-drive system based on an in-vehicle tamper-proof device. Others adopt blockchain-based decentralized approaches [5], [6], where drivers periodically upload encrypted driving data (e.g., speed, acceleration, time) to the blockchain. This data is used to compute insurance premiums via matrix encryption. While these schemes avoid trusted third parties, they incur high communication and storage costs and require AI integration to personalize billing based on environment and user behavior.

Beyond cryptography, AI techniques such as Autoencoders [7] and Knowledge Distillation [8] offer confidentiality benefits. In Autoencoder-based schemes, each client trains a local encoder to produce latent representations, which are shared with a server for model training without exposing raw data. Distillation techniques enable a smaller student model to mimic a larger teacher model using response, feature, and relation-based knowledge. However, these approaches lack formal security guarantees and remain vulnerable to attacks like gradient inversion and membership inference [20]–[23].

In terms of cryptographic ML construction, secret sharing-based approaches [9]–[11] rely on MPC protocols such as Arithmetic, Binary, and Yao (ABY). A typical scenario involves three-party computation with oblivious transfers, enabling secure model training. Despite good privacy guarantees, such systems suffer from high communication overhead, limited scalability, and long execution times.

Homomorphic Encryption (HE) enables encrypted data processing and is often combined with MPC [12]–[15], [24]. In these schemes, clients hold partial keys and jointly decrypt results using threshold cryptography. The Pivot framework [12], for example, applies HE+MPC to train tree-based models (e.g., GBDT) in vertical federated learning. It assumes honest-but-curious participants and avoids trusted third parties. However, its security weakens with few participants, and real-time use is impractical due to communication complexity and exponential growth

TABLE I
COMPARISON OF THE LIMITATIONS ADDRESSED BY THE RELATED WORK AND THE PROPOSED APPROACH, GI IS THE NUMBER OF GLOBAL ITERATIONS

Research	Privacy Preserving for UBI	Protocol	ML Based	Scalability by design	Number of communications	Feasibility
[4]–[6]	Medium	Blockchain	✗	Medium	–	✓
[7], [8]	Low	Autoencoder/Distillation	✓	High	O(GI)	✓
[9]–[11]	High	ABY/Secret Sharing	✓	Low	O(2*GI)	✗
[12]–[15]	Medium	HE + MPC	✓	High	O(GI)	✓
[16]–[19]	Medium	DP	✓	High	O(GI)	✓
Our Work	High	HE + DP	✓	High	O(GI)	✓

in computation as tree depth increases. Further efforts [25] address these issues in private UBI but often assume a fully trusted key manager.

III. PRELIMINARIES

A. Differential Privacy (DP)

This section begins by introducing essential concepts. Differential Privacy (DP), as defined in [26], offers a rigorous framework for releasing insights derived from sensitive data. When applied to machine learning, DP enables the public sharing of model parameters while providing strong privacy guarantees. Specifically, it limits what adversaries, regardless of their access to auxiliary information, can infer about any individual record in the training data. Formally, DP is defined as follows:

Definition III.1. A randomized mechanism $M : D \rightarrow R$ with domain D (e.g., possible training datasets) and range R (e.g., all possible trained models) satisfies (ϵ, δ) -differential privacy if, for all pairs of **adjacent** datasets $d, d' \in D$ and for all subsets of outputs $S \subseteq R$, it holds that:

$$\Pr[M(d) \in S] \leq e^\epsilon \Pr[M(d') \in S] + \delta$$

Here, (ϵ, δ) quantify the privacy loss: ϵ bounds the worst-case leakage, while δ denotes the probability that this bound is violated (ideally $\delta < 1/|d|$).

To enforce (ϵ, δ) -differential privacy, one typically adds carefully calibrated noise to the output of a function, proportional to its sensitivity. Common noise-adding mechanisms include the Gaussian, Laplace, and exponential mechanisms, each chosen based on the desired privacy-utility tradeoff.

B. Applying DP in FL

In this work, we use the approach based on the DP-AVG algorithms from the paper [27]. To implement DP-AVG, we separate each batch into micro-batches

of size one, compute the gradients on these micro-batches, clip, and add noise: 1) For batching, we use a uniform sampling of batches (also called Poisson sampling). Poisson sampling enables each data point to be independently added to the batch with a probability equal to the sampling rate. 2) During the implementation, DP-AVG needs to compute clipped gradients. Clipping is a way of bounding gradients. If its norm is larger than the target bound for the maximum gradient norm, called C . The gradient must be clipped by scaling it down to C . 3) The noise added is calibrated to $M(d)$'s sensitivity ($S_{M(d)}$). $S_{M(d)}$ defines the maximum of the absolute distance $|M(d) - M(d')|$ where d and d' are adjacent inputs. We use the ‘‘Gaussian mechanism’’ which is defined by $\mathcal{M}_{Gauss}(d) = M(d) + \mathcal{N}(0, S_{M(d)}^2 \cdot \sigma^2)$. where $\mathcal{N}(0, S_{M(d)}^2 \cdot \sigma^2)$ is the normal distribution with the mean 0 and standard deviation $S_{M(d)}^2 \cdot \sigma^2$. The error is computed using functions from the Opacus library developed by Meta AI [28]. As presented in the paper [29] Theorem 3.22, $M(d)$ satisfied (ϵ, δ) -DP if $\delta \geq \frac{4}{5} \exp(-(\sigma\epsilon)^2/2)$ and $\epsilon < 1$.

The DP mechanism is constructed of one algorithm, named *AddNoise*:

- *AddNoise*(ω_E^t). For given $S_{M(d)}$ L2-sensitivity of $M(\omega_E^t, D_t)$ in the t -th GI. An output is the Gaussian noise $e_t \simeq \mathcal{N}(0, S_{M(d)}^2 \cdot \sigma^2)$. It determines to which noise is added (How much noise to add).

IV. PRIVFEDPROFILING: PRIVACY-PRESERVING FEDERATED PROFILING APPROACH

A. PrivFedProfiling: Architecture, and Threat Model

This work addresses the secure decentralized training of machine learning (ML) classifiers using sensitive data collected from vehicles in the context of Usage-Based Insurance (UBI). We focus on the training phase and present the PrivFedProfiling system, designed to protect driver privacy while building

an accurate global driving behavior classifier. As presented in Table II, the vehicles are classified into three categories: *slow*, *normal*, and *dangerous*, based on behavioral features such as speed, braking, and acceleration.

TABLE II
BREAKDOWN OF DRIVING BEHAVIOURS

Type	Definition	Bound
Slow	Indicating the maintenance of a speed lower than the average.	$[0.33, 1]$
Normal	Representing average driving events.	$[-0.33, 0.33]$
Dangerous	Characterized by sudden acceleration, braking, and aggressive behavior.	$[-1, -0.33]$

System Model. The system is composed of three main entities, as shown in Fig. 2: • **Central Server (CS)** selects labeled vehicles to participate in the training phase, initializes the global model with synthetic data (e.g., from SUMO), distributes it to vehicles, and updates it using aggregated results. • **Aggregator Server (AS)** is a high-performance server that performs encrypted aggregation of local models. • **Vehicles ($\mathcal{V}_t$)** hold sensitive real-world data. Using the global model and labels from CS, each vehicle trains a local model, which is encrypted and sent to AS.

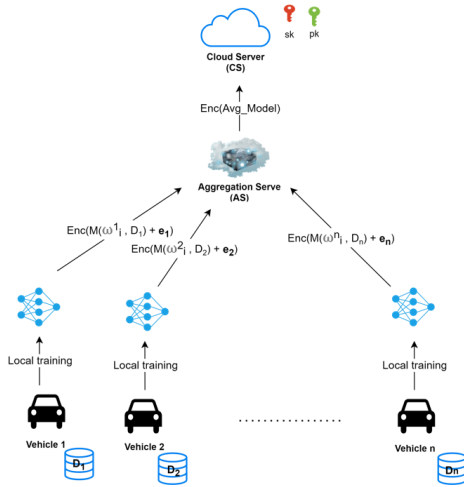


Fig. 2. PrivFedProfiling Architecture and HE key distribution

Threat Model and Privacy Goals. We assume all parties ($\mathcal{V}_t$, CS, AS) are *honest-but-curious*. They follow the protocol but may attempt to infer sensitive data. Our goals are:

- 1) CS may attempt to extract sensitive data from decrypted aggregates, such as speed or location.
- 2) Vehicles are assumed to have honestly trained their local models. The FL process still functions correctly without the privacy layer, but privacy risks remain.
- 3) CS may collude with one or more vehicles to infer private details.
- 4) AS may try to learn from the encrypted models, but is limited to aggregation operations.
- 5) CS and AS do not collude.

Protocol Overview. The PrivFedProfiling model (Fig. 3) operates in multiple steps. First, the central server CS trains a global model using synthetic labeled data generated from SUMO. Then, each vehicle $\mathcal{V}_t$ utilizes the global model and its associated labels to train a local model on its own private data. These locally trained models are subsequently encrypted and transmitted to the aggregation server AS for secure aggregation. Once aggregated, the central server CS decrypts the result and updates the global model accordingly. This process is repeated, excluding the initial synthetic training step, until the global model reaches convergence.

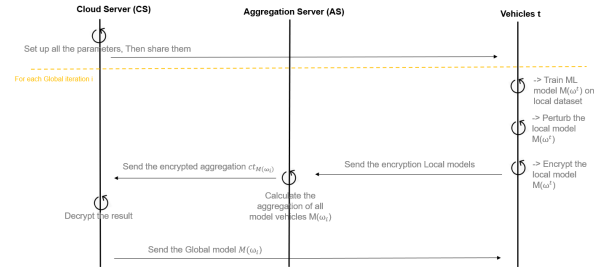


Fig. 3. Illustration of PrivFedProfiling protocol

B. PrivFedProfiling Model Construction

The description of federated learning is detailed in Section III-B. This section is dedicated to formally describing our construction, which applies HE and DP to the UBI scenario and satisfies all the requirements defined in Section IV. Our goal is to create a global model by incorporating data from all vehicles while preserving data privacy. The construction is based on the following three algorithms:

- **VehTrain**($\mathcal{D}_t, M(\omega_i)$) presented in Algorithm 1 is executed by each vehicle t , where $\mathcal{D}_t$ is the local dataset. Parameters: group size $\mathcal{B}$, learning rate η_t , noise scale σ , gradient norm bound $\mathcal{C}$, and the cloud public key pk . The dataset is split into batches of size $\mathcal{B}$. Each vehicle $\mathcal{V}_t$

retrieves the model parameters ω from the cloud service (CS), computes gradients using a loss function $f()$, clips the ℓ_2 norm of each gradient (as described in Section III-B), computes the average, adds noise for privacy, and encrypts the model using CS 's public key pk (as described in paper [30]). The algorithm returns an encrypted noisy model $ct_{M(\omega^t)}$.

- **SecureAgg**($ct_{M(\omega^0)}, \dots, ct_{M(\omega^{n-1})}, n$) presented in Algorithm 2 is executed by the aggregation server (AS). It sums all encrypted models and multiplies the result by the inverse of the number of vehicles n , because the division is not supported in SHE. The algorithm returns an encrypted noisy model $ct_{M(\omega_i)}$ for the i -th global iteration (GI).
- **CloudUpdate**(GI) presented in Algorithm 3 is run by the cloud service provider CS , which orchestrates the entire protocol. The algorithms VehTrain and SecureAgg are executed outside the cloud. The cloud initializes the ML model ω_0 and trains it using a dataset from a simulator (e.g., SUMO). It selects a SHE environment (n, q, t, σ) and generates HE keys using $KeyGen(1^\lambda)$ as described in the paper [30]. It then selects n vehicles for training, sends them the public parameters, and waits for the encrypted aggregation result $ct_{M(\omega_i)}$ from AS . This result is decrypted using Dec_{sk} and sent back to the vehicles for the next GI.

V. IMPLEMENTATION AND EVALUATION

In this section, we present a comprehensive experimental evaluation using datasets generated by the SUMO simulator to validate our theoretical findings.

A. Database Description

Due to the sensitivity of driver data, CS must not have direct access to it. The recommended approach for this use case begins by initializing the ML training phase with simulated data. Subsequently, the PrivFedProfiling algorithm is employed to refine and personalize the model with real-world driver data. In this study, we utilize only simulated data across all protocols, as real-world data is currently unavailable.

We generate datasets using the SUMO simulator, categorizing drivers into three behavior profiles: Slow, Normal, and Dangerous. Modern automotive technologies—such as Automatic Emergency Braking (AEB), Advanced Driver Assistance Systems (ADAS), Lane-Keeping Assist (LKA), and Blind

VehTrain ($\mathcal{D}_t, M(\omega_i), pk, \eta, \sigma, \mathcal{C}, Id_{veh} = t$)

```

 $\mathcal{K} \leftarrow$  Split data  $\mathcal{D}_t = \{x_1, \dots, x_N\}$  into
batches of size  $\mathcal{B}$  with sampling
probability  $\mathcal{B}/\|\mathcal{D}_t\|$ 
Initialize  $\omega_0 \leftarrow \omega_i$ 
for  $k$  in  $\mathcal{K}$  do
  Compute gradient:
  For each  $i \in \mathcal{B}_t$ , compute
   $\mathbf{g}_k(x_i) \leftarrow \nabla_{\omega_k} f_i(\omega_k, x_i)$ 
  Clip gradient:
   $\hat{\mathbf{g}}_k(x_i) \leftarrow \mathbf{g}_k(x_i) / \max(1, \frac{\|\mathbf{g}_k(x_i)\|_2}{\mathcal{C}})$ 
  Add noise:
   $\bar{\mathbf{g}}_k \leftarrow \frac{1}{\mathcal{B}} (\sum_i \hat{\mathbf{g}}_k(x_i) + \mathcal{N}(0, \sigma^2 \mathcal{C}^2 \mathbf{I}))$ 
  Gradient descent:
   $\omega_{k+1} \leftarrow \omega_k - \eta \omega \bar{\mathbf{g}}_k$ 
end
 $\omega^t \leftarrow \omega_{k+1}$ 
for  $key, value$  in  $M(\omega^t)$  do
  |  $ct_{M(\omega^t)}[key] \leftarrow Enc_{pk}(value)$ 
end
return  $ct_{M(\omega^t)}$ 

```

Algorithm 1: The local learning algorithm executed by each vehicle.

SecureAgg ($ct_{M(\omega^0)}, \dots, ct_{M(\omega^{n-1})}, n$)

```

|  $ct_{M(\omega_i)} \leftarrow \frac{1}{n} \sum_{t=0}^{n-1} ct_{M(\omega^t)}$ 
return  $ct_{M(\omega_i)}$ 

```

Algorithm 2: The algorithm run by AS on encrypted local models.

CloudUpdate (GI)

```

( $sk, pk$ )  $\leftarrow GenKey(1^\lambda)$ 
Select  $n$  vehicles for training
Initialize  $\omega_0$  with dataset from SUMO
simulator
for  $i \leftarrow 0$  to  $GI-1$  do
  for  $t \leftarrow 0$  to  $n-1$  do
    |  $ct_{M(\omega_i^t)} \leftarrow$ 
    |  $VehTrain(\mathcal{D}^t, M(\omega_i), E^t, pk)$ 
  end
   $ct_{M(\omega_i)} \leftarrow$ 
   $SecureAgg(ct_{M(\omega_i^0)}, \dots, ct_{M(\omega_i^{n-1})}, n)$ 
   $M(\omega_i) \leftarrow Dec_{sk}(ct_{M(\omega_i)}, sk)$ 
end
return  $M(\omega_{GI-1})$ 

```

Algorithm 3: The overview of the PrivFedProfiling algorithm executed by CS .

Spot Detection (BSD)—provide a range of warnings, including visual, audible, haptic, and head-up display

notifications, with some systems taking control of the vehicle. These mechanisms serve as classification alternatives to telematics data.

To simulate comparable warnings, we apply the AlertDang Driver Profiling method and SUMO parameters introduced in [31]. The generated dataset includes five types of alerts inspired by these technologies (see more in [31]): • **Speed Limit Respect**: Compares vehicle speed to lane-specific limits. • **Front Safe Distance**: Ensures a minimum inter-vehicular gap of 2.5 meters. • **Secure Distance**: Computes a time-based safe following distance, with alerts triggered when this falls below 1.5 seconds. • **Time To Collision (TTC)**: Issues a warning when TTC is below 2 seconds. • **Emergency Brake**: Triggers when sudden deceleration exceeds expected levels.

We evaluate the performance of *PrivFedProfiling* using datasets generated on a simulation of the Miami city map, which includes various road types and traffic signs. The source code and datasets are available on GitHub⁴. Our implementation is adaptable to any dataset. We collect data for both *CS* and the individual vehicle datasets $\mathcal{D}_t$. The initial dataset for *CS* includes 3,000 trips (1,050 per driver category), totaling 1,077,746 samples: 413,421 slow, 346,565 normal, and 317,760 dangerous samples. Additionally, we generate 88 individual datasets $\mathcal{D}_t$ for 88 vehicles: 29 Slow, 29 Normal, and 30 Dangerous. Each $\mathcal{D}_t$ contains between 90 and 120 trips, consistent with the monthly average of 90 trips per driver [32]. Altogether, we simulate 14,480 trips (3,400 per driver category), resulting in 8,171,462 samples: 1,996,659 slow, 1,653,936 normal, and 1,443,331 dangerous.

B. Experimental Settings

All experiments are executed on a system equipped with a 12th Gen Intel(R) Core(TM) i9-12950HX processor running at 2.30 GHz and 32 GB of RAM. We employ a Multilayer Perceptron (MLP) model for classification, though the approach is agnostic to the choice of ML algorithm regarding HE and DP performance. The MLP is configured with hidden layers of sizes (4, 8, 4), mean squared error (MSE) as the loss function, ‘relu’ as the hidden layer activation function, ‘adam’ as the optimizer, and learning rates of 10^{-5} and 10^{-4} . We also apply the hyperbolic tangent (tanh) activation for output normalization in the range [-1, 1], corresponding to the driver behavior classes: Slow ([-1, -0.33]), Normal ([-0.33, 0.33]), and Dangerous ([0.33, 1]).

For privacy, we configure the differential privacy parameter ϵ to values 0.2, 0.3, and 0.4. A smaller ϵ provides stronger privacy but introduces more noise. The number of learning iterations *LI* is set to 1. We use the **TenSEAL** library to implement homomorphic encryption, configuring the CKKS scheme with a polynomial modulus degree of 8192 and coefficient modulus bit sizes of [60, 40, 40, 60] to balance precision and security. The global scale is set to 2^{40} for precise computation.

We evaluate two learning structures:

- **NoDP-HE**: Federated learning without privacy-preserving mechanisms.
- **PrivFedProfiling**: Federated learning integrating both Homomorphic Encryption (HE) and Differential Privacy (DP).

C. PrivFedProfiling Performance Analysis

For UBI services, we compare the two federated learning structures described above. Table III presents the performance metrics for the operations defined in Section IV-B: *Setup()*, *VehTrain()*, *SecureAggr()*, and *CloudUpdate()*.

In each experiment, we simulate 15 vehicles selected randomly for each global iteration, one *CS*, and one *AS*. The cloud *Setup()* phase takes approximately 235 ms due to initialization of the MLP model, Poisson sampling, CKKS scheme, and DP mechanism, with the CKKS setup being the most time-consuming. Each vehicle’s *Setup()* takes about 40 ms to initialize the same components locally.

The local training phase, *VehTrain()*, takes approximately 0.35 seconds per vehicle. This time is constant regardless of the number of vehicles, as training is performed independently.

The *SecureAggr()* phase depends on the number of vehicles n and the number of model layers h . Each vehicle sends encrypted model weights and biases, totaling $2h$ ciphertexts per vehicle. Aggregation adds $(n - 1) \cdot 2h$ ciphertexts and performs $2h$ homomorphic multiplications. With $n = 15$ and $h = 4$, this operation takes around 35 ms. We assume that one homomorphic addition takes roughly 0.0001 ms, and one homomorphic multiplication takes approximately 2 ms. Based on this analysis, the *PrivFedProfiling* protocol demonstrates strong scalability and efficient performance under privacy constraints.

D. PrivFedProfiling Accuracy Analysis

We conduct two simulations to evaluate the accuracy of *PrivFedProfiling*. In the first, a global model

TABLE III
COMPARISON OF PERFORMANCE WITH AND WITHOUT HE AND DP

Algorithm	Cloud Setup (ms)	Vehicle Setup (ms)	VehTrain (s)	SecureAggr (ms)	CloudUpdate (ms)
PrivFedProfiling	235	≈ 170	≈ 0.35	≈ 35	≈ 18
NoDP-HE	2	≈ 65	≈ 0.35	0	≈ 0

TABLE IV
IMPACT OF PRIVACY BUDGET ON ACCURACY. ¹: WITH CLOUD PRE-TRAINING; ²: VEHICLE-ONLY TRAINING

Algorithm	ϵ	Accuracy	GI	η
NoDP-HE	1	99%	1	10^{-5}
PrivFedProfiling ¹	0.2/0.3/0.4	98%	1	10^{-5}
PrivFedProfiling ²	0.4	98%	3	10^{-5}
PrivFedProfiling ²	0.3	93%	4	10^{-5}
PrivFedProfiling ²	0.2	94%	7	10^{-5}
PrivFedProfiling ²	0.2/0.3/0.4	99%	1	10^{-4}

is pre-trained using cloud data, resulting in poor accuracy on vehicle data. After one global epoch, the privacy budget ϵ does not affect accuracy, but further evolution cannot be observed due to the single iteration. The second simulation removes the initial cloud-based training. Vehicles collaboratively construct the model from scratch. As the privacy budget decreases (i.e., stronger privacy), model accuracy declines, and more global iterations (GIs) are needed. For $\epsilon = 0.4, 0.3$, and 0.2 , the model converges in 3, 4, and 7 GIs, respectively, with accuracies of 98%, 93%, and 94%.

Figures 4, 5, and 6 show the evolution of local model accuracy and loss. We compute the average accuracy per vehicle type (slow, normal, dangerous). The use of tanh activation (with labels mapped to -1, 0, 1) improves convergence. Slow vehicles achieve better performance than dangerous ones, likely due to data overlaps and label ambiguity.

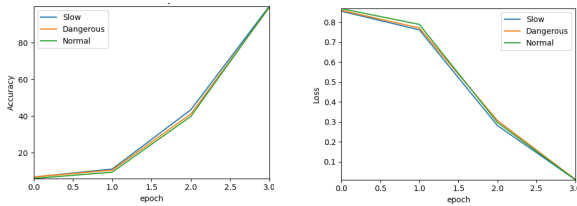


Fig. 4. PrivFedProfiling² accuracy evaluation. $\epsilon = 0.4$ and $\eta = 10^{-5}$.

Impact of Privacy Budget and Learning Rate.

The homomorphic encryption layer (CKKS) does not affect accuracy or GIs. However, the DP layer impacts accuracy depending on the value of ϵ and whether cloud-based pretraining is used. When vehicles train from scratch, low ϵ values increase GIs and reduce accuracy. Raising the learning rate to $\eta = 10^{-4}$

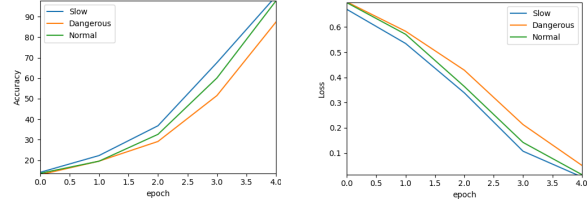


Fig. 5. PrivFedProfiling² accuracy evaluation. $\epsilon = 0.3$ and $\eta = 10^{-5}$.

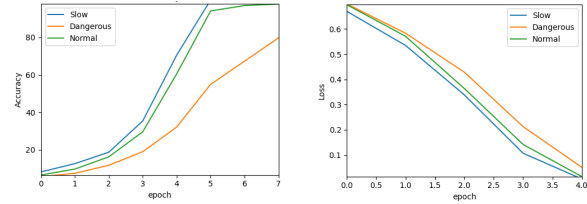


Fig. 6. PrivFedProfiling² accuracy evaluation. $\epsilon = 0.2$ and $\eta = 10^{-5}$.

compensates for this effect, enabling convergence in one GI and restoring accuracy to 99% (Table IV).

Security Analysis and Privacy Requirements

The proposed *PrivFedProfiling* approach ensures strong privacy guarantees based on both DP and HE. The privacy requirements satisfied by the system can be summarized as follows:

- **Differential Privacy Guarantee:** Each vehicle $\mathcal{V}_t$ trains a local model with (ϵ_t, δ) -differential privacy using the Opacus library [28]. The aggregation mechanism ensures that the final global model satisfies an average privacy budget $\epsilon = \frac{\sum(\epsilon_1, \dots, \epsilon_n)}{n}$ as established by the composition property of differentially private mechanisms [27], [33].
- **Privacy against Central Server (CS):** The local models remain indistinguishable to the central server CS, even in cases where CS collaborates with a subset of vehicles or the aggregator server (AS). This ensures privacy preservation for non-colluding vehicles, fulfilling the *Privacy-Preserving Data Owner* requirement.
- **Privacy against Aggregator Server (AS) and Other Vehicles:** The local models are en-

encrypted using the CKKS homomorphic encryption scheme, which is CPA-secure under the Ring Learning With Errors (R-LWE) assumption [30], [34]. This ensures that $\mathcal{AS}$ and any colluding vehicles cannot infer any sensitive information from the encrypted updates.

VI. CONCLUSION

We introduced a privacy-preserving scheme for UBI services using federated learning, homomorphic encryption, and differential privacy. Leveraging SUMO-generated warnings, our method enables real-time, scalable driver profiling with minimal performance loss, even at high privacy levels ($\epsilon = 0.2$, $\eta = 10^{-5}$). The code and datasets are available on GitHub. Future work includes improving modularity, testing scalability, and integrating real drivers to explore social and ethical impacts.

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